

VISIONLOT

Dealership Parking Occupancy Detector

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ITAI 1378

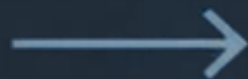
Tier 1



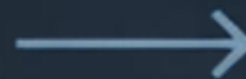
THE PROBLEM

Outdated parking records
create uncertainty and
waste valuable time.

Location
recorded



Vehicle
moved



Employees
search the lot



THE PROPOSED SOLUTION

VisionLot uses computer vision to monitor 10–20 clearly defined parking spaces from a photograph and report occupancy.

01 → 02 → 03 → 04



SEVENTH-FLOOR PARKING VIEW

A photo captures the lot from the seventh floor.



YOLO11 DETECTION

YOLO11 detects vehicles in the photograph.



PREDEFINED SPACES

Spaces are mapped and classified as occupied or vacant.

OCCUPANCY REPORT

OCCUPIED

13

VACANT

7

OCCUPANCY REPORT

Occupancy report shows occupancy and vacancy status.

Statuses shown are illustrative.

TECHNICAL APPROACH

YOLO11

detects vehicles in the image



Python + OpenCV

match detections to parking spaces



Parking space regions

define occupancy per spot



Google Colab

runs the proof of concept



DATA PLAN

STEP 1

3-5

initial sample
images

STEP 2

30-50

planned image
captures

STEP 3

10-20

visible spaces
monitored

**LABEL
QUALITY**

each space manually checked as
occupied or **vacant**



SUCCESS METRICS

$\geq 90\%$

parking-space
occupancy accuracy
minimum target

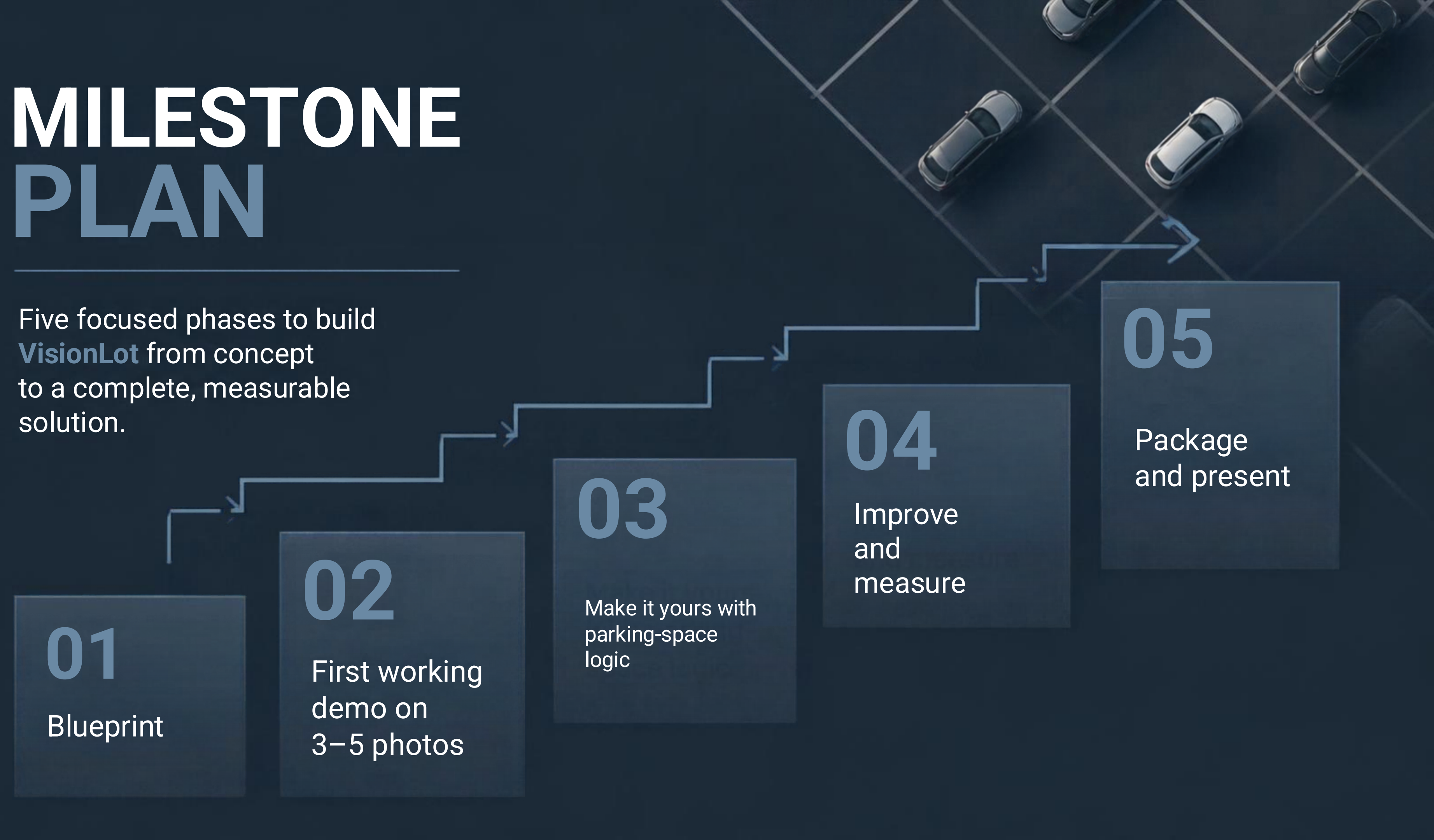
≤ 2 sec

processing time
per image
maximum target
in Google Colab

Proposed targets. No measured results yet.



MILESTONE PLAN



```
graph LR; 01[01 Blueprint] --> 02[02 First working demo on 3-5 photos]; 02 --> 03[03 Make it yours with parking-space logic]; 03 --> 04[04 Improve and measure]; 04 --> 05[05 Package and present];
```

Five focused phases to build **VisionLot** from concept to a complete, measurable solution.

01

Blueprint

02

First working
demo on
3–5 photos

03

Make it yours with
parking-space
logic

04

Improve
and
measure

05

Package
and present

RISKS AND RESOURCES

KEY RISKS

Small or missed vehicles
Camera shifts

Crop to clear spaces
and test threshold

Consistent viewpoint
and alignment

TOOLS

Smartphone
Python
YOLO11
OpenCV
Google Colab

ESTIMATED COST

\$0

Using existing equipment
and free compute

